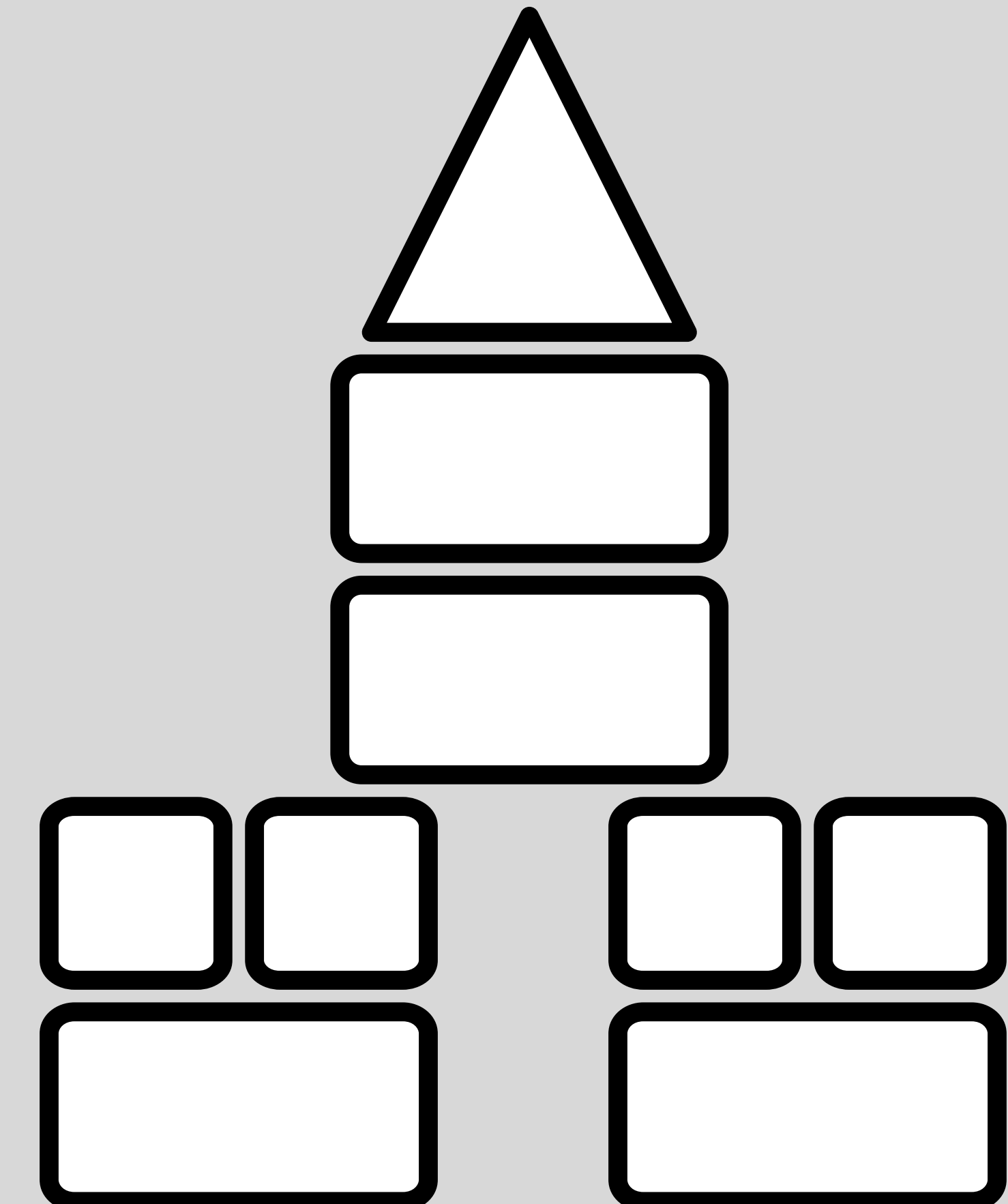
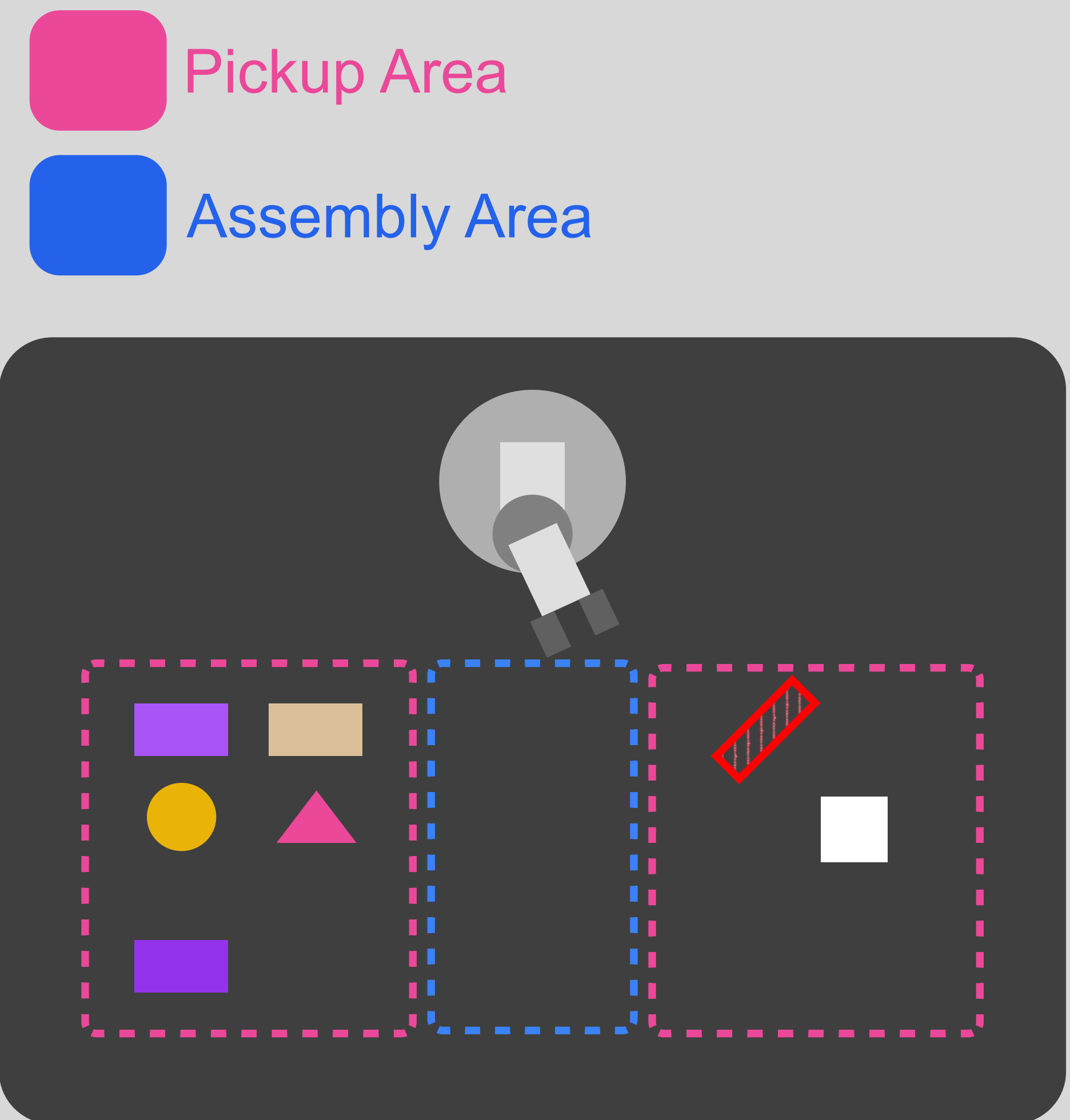


Input

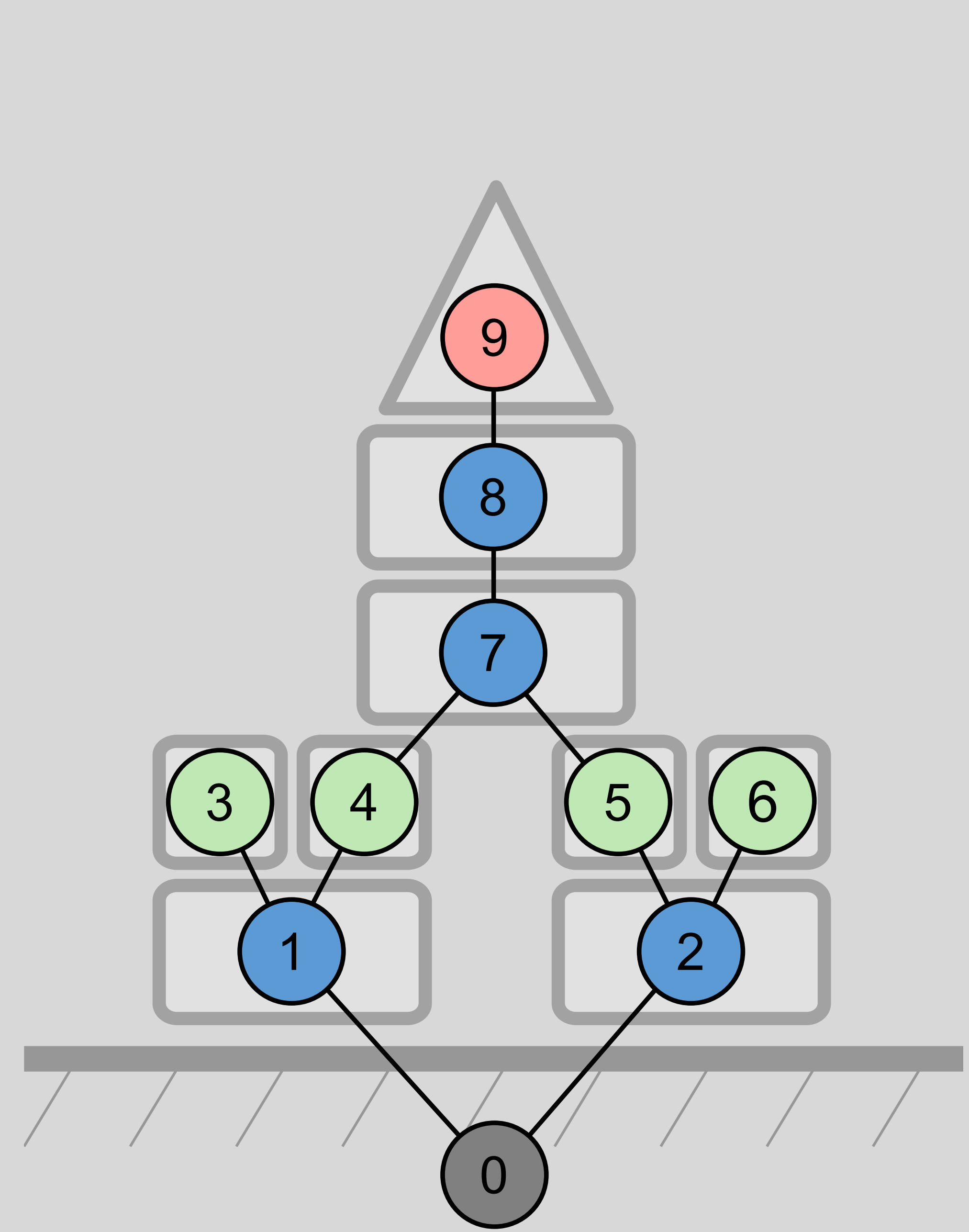


Task Specification

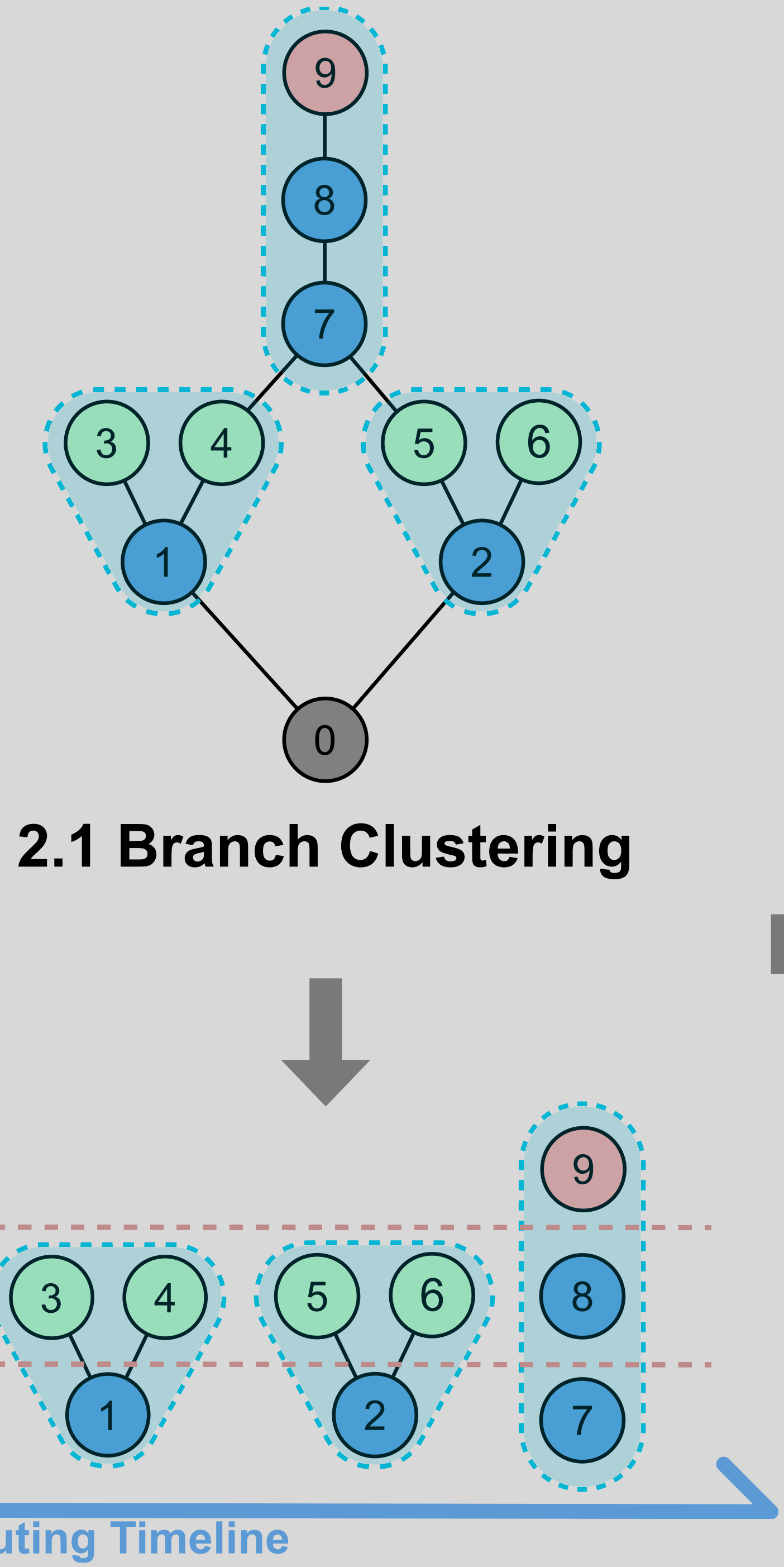


Initial Scene

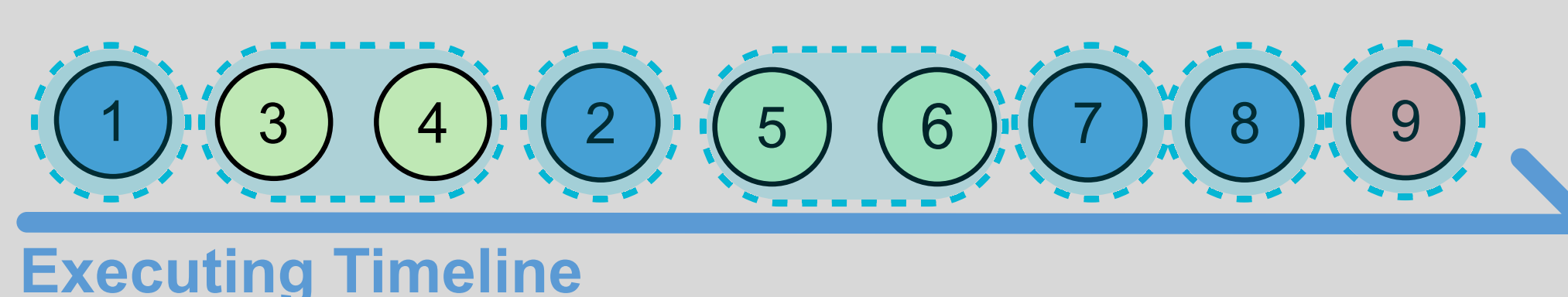
Planning



1. VLM Graph Generation



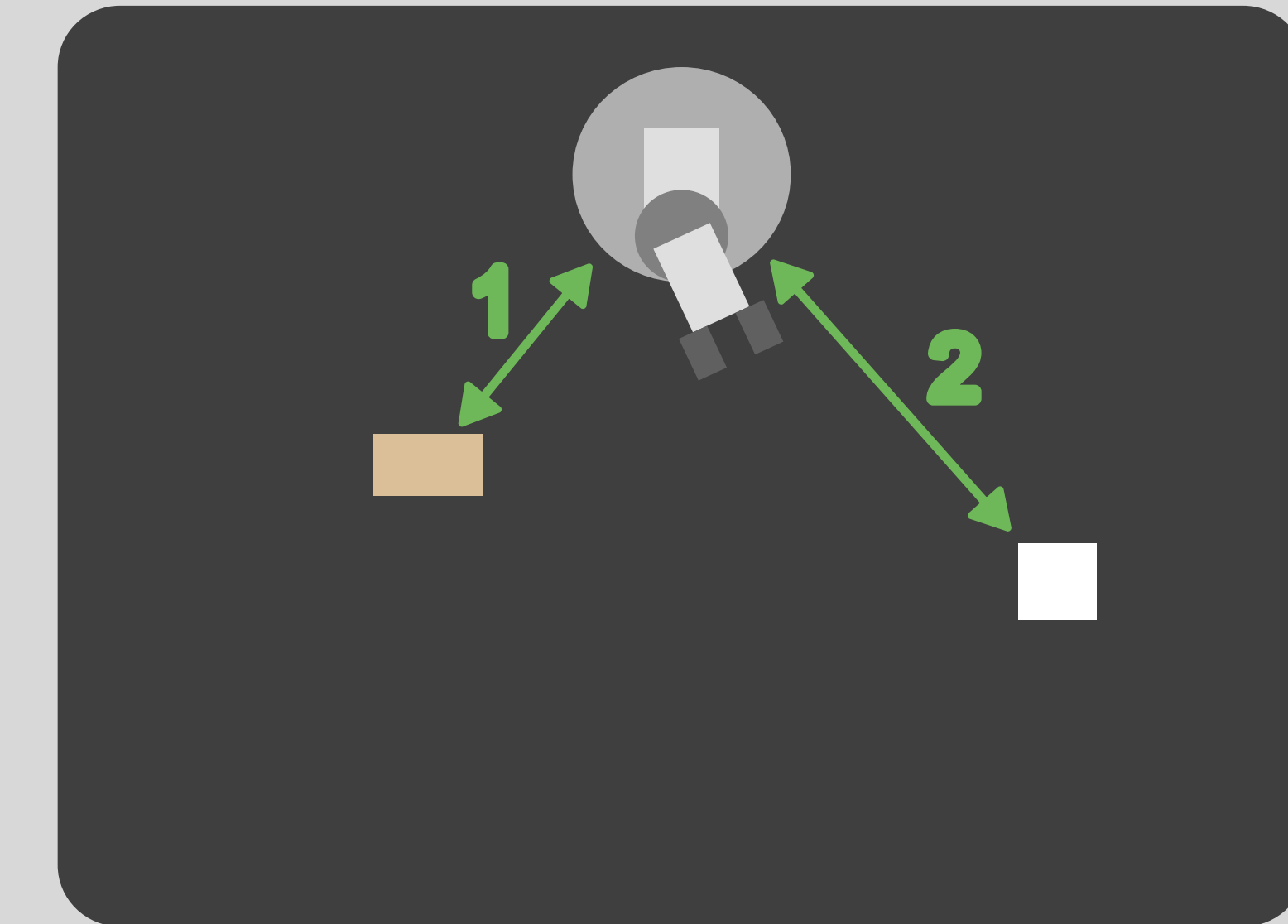
2. Two-Level Clustering



3. Global Execution Sequence

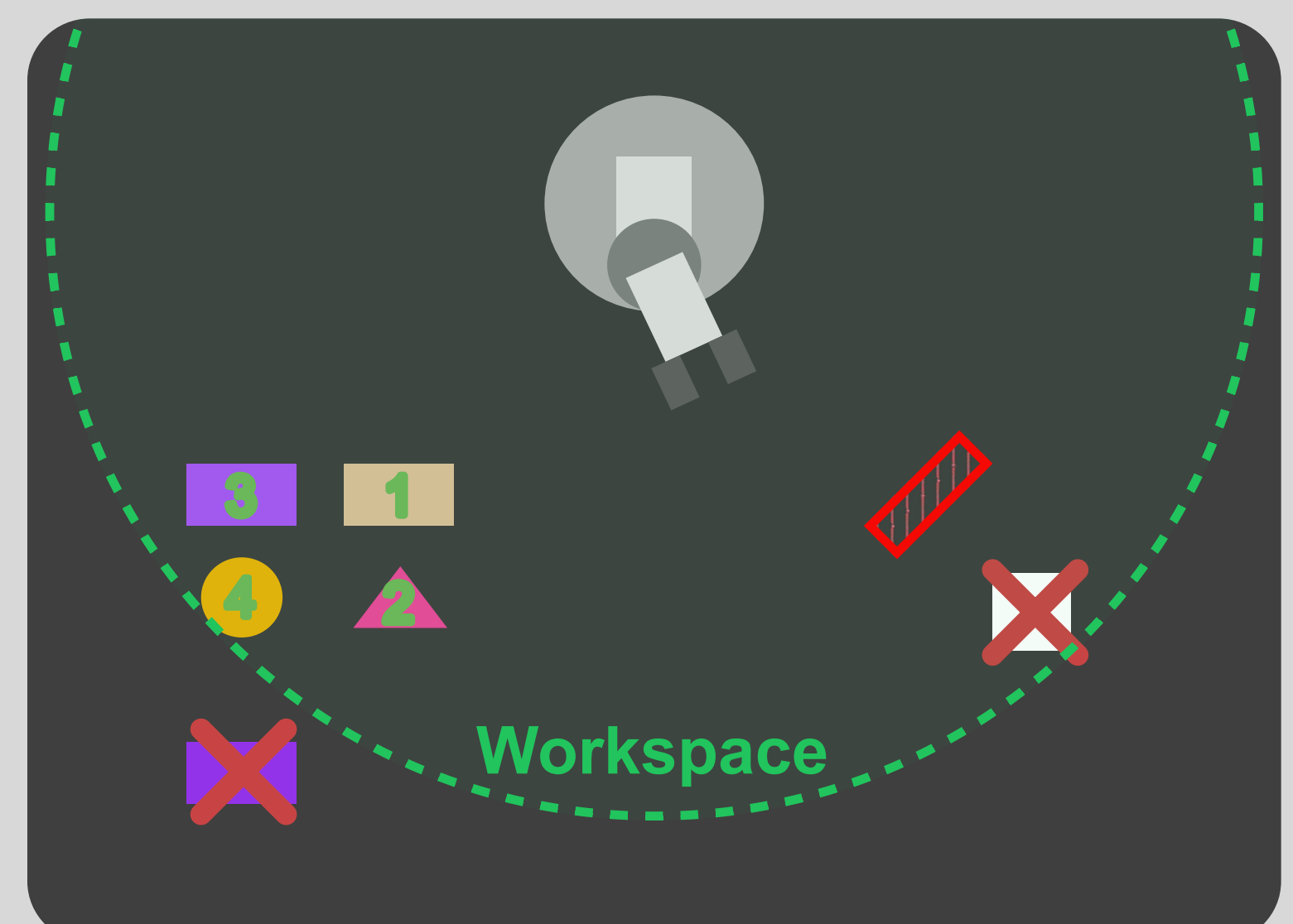


0.1.1 Reachability Detection

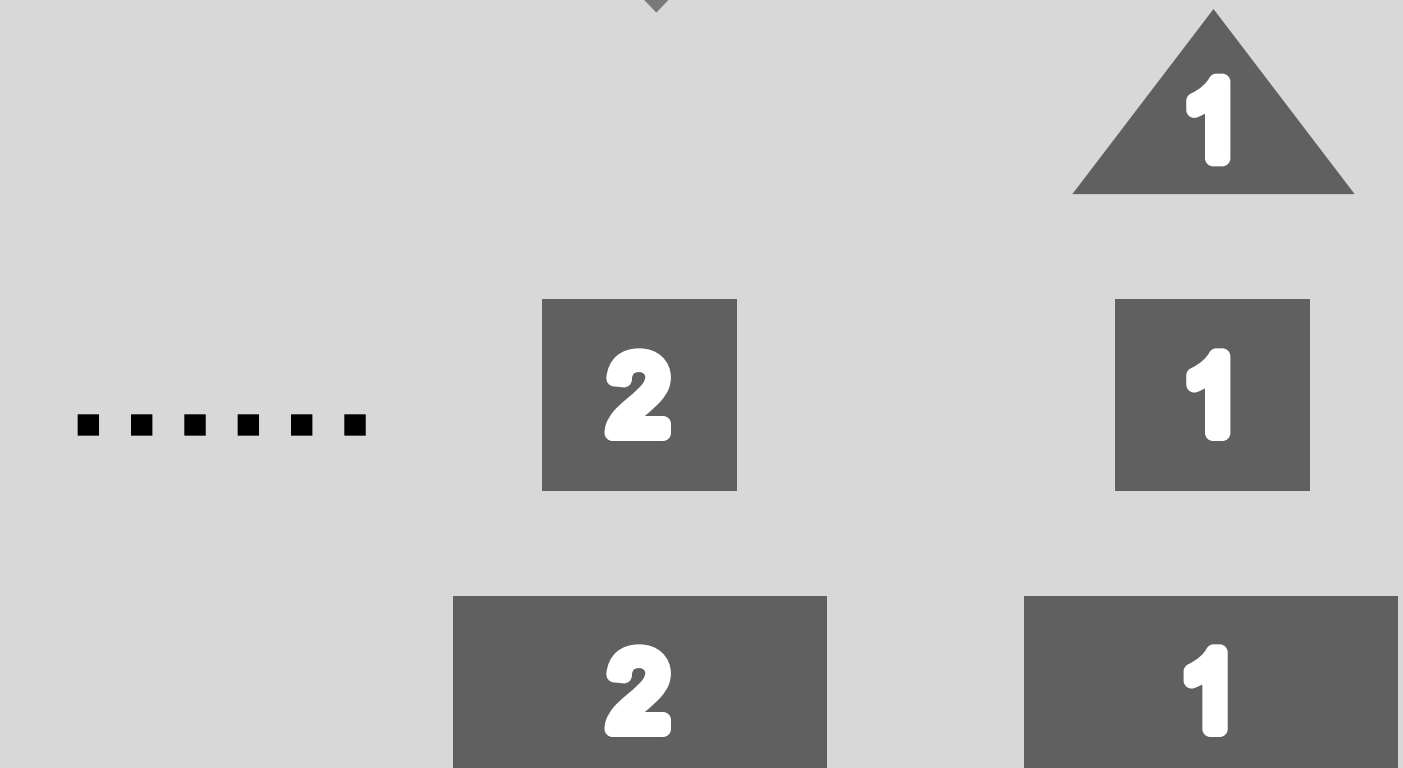


0.1.2 Euclidean Distance Detection

0.1 Workspace Pruning & Ordering



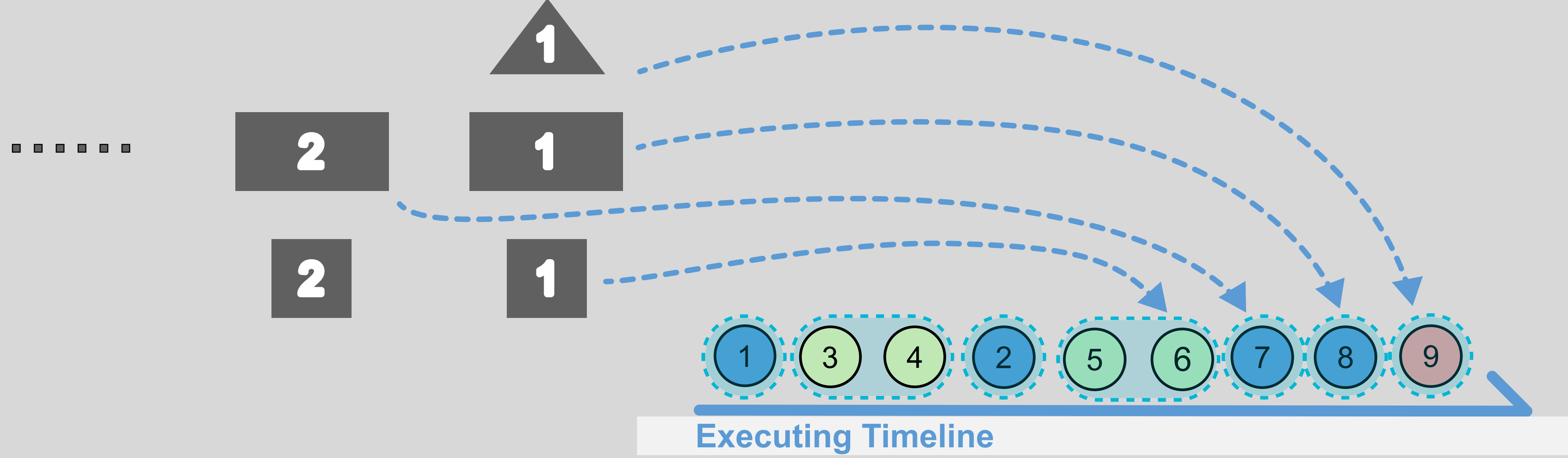
0.2.1 Pruned by reachability,



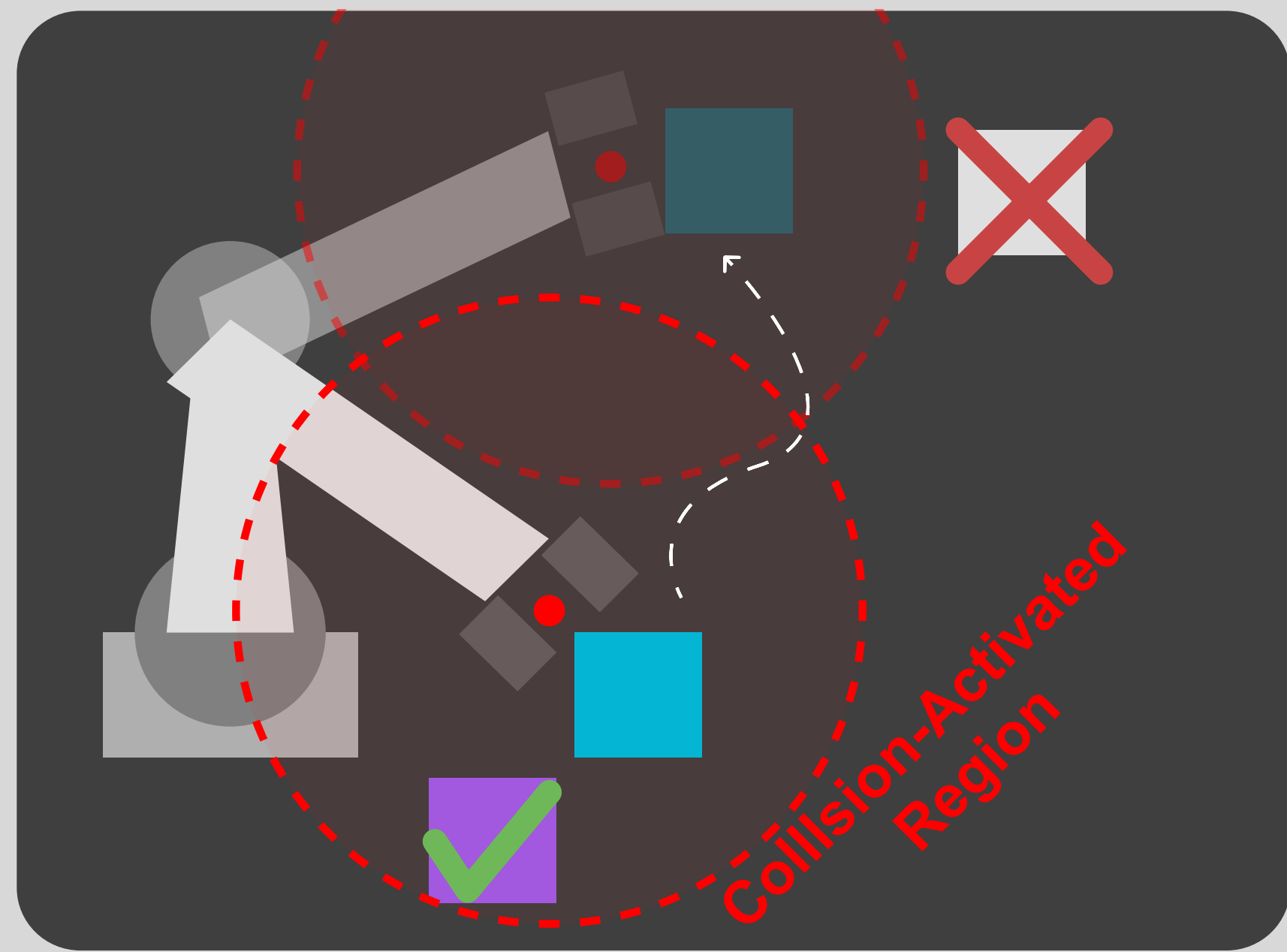
0.2.2 Ordered by Euclidean distance

0.2 Scene Grounding

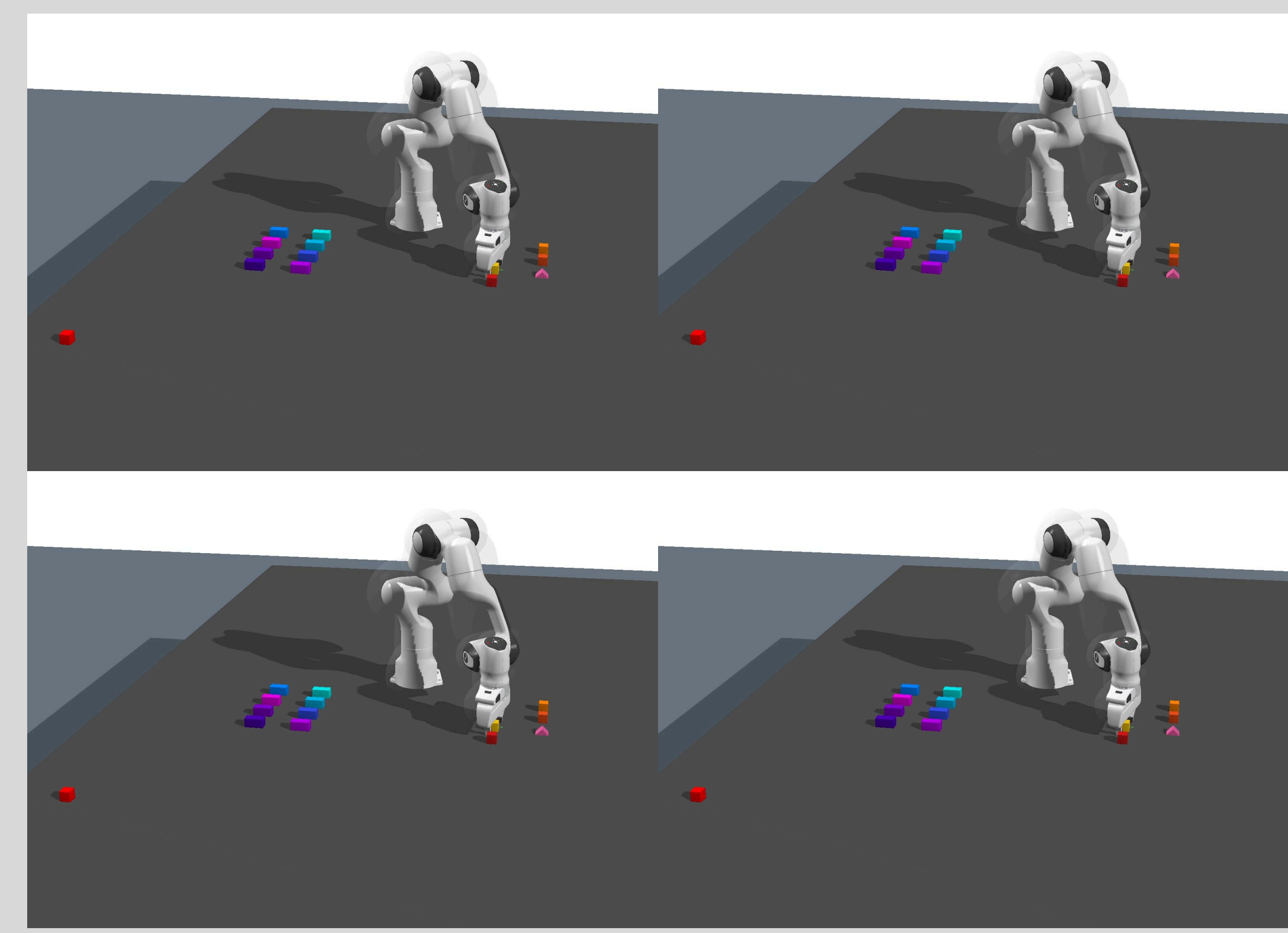
Solving & Executing



Execution-Order Embedding



Selective LGP Solving with Active Constraints



Real Robot Execution